

A Hybrid Method for On-line Performance Assessment and Life Prediction in Drilling Operations

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Abstract - Tool wear condition monitoring and remaining life prediction are critical for near-zero downtime machining. Recent manufacturing outsourcing business environment necessitates more focus on machine performance degradation to optimize the tool management for improved six-sigma productivity and manufacturing performance. The unmet needs for drilling monitoring is how to effectively predict its remaining life and manage the tool change to minimize downtime and costs.

This paper presents a hybrid method for on-line assessment and performance prediction of remaining tool life in drilling operations based on the vibration signals. Logistic regression (LR) analysis combined with maximum likelihood technique is employed to evaluate tool wear condition based on features extracted from vibration signals using Wavelet Packet Decomposition (WPD) technique. Auto-regressive Moving Average (ARMA) model is then applied to predict remaining useful life based on tool wear assessment result. In addition, failure risk distribution is discussed. The developed prognostic method is validated in drilling operations, which can be also implemented to other manufacturing processes.

Index Terms - Condition monitoring; remaining life prediction; prognostics; tool wear; drilling monitoring.

I. INTRODUCTION

On-line tool wear monitoring and predictive tool replacement at the proper time are important techniques to realize a fully automated manufacturing system and prevent machine downtime. Drilling is a major material removal processes in manufacturing, it represents approximately 40% of all cutting operations performed in industry [1]. Therefore,

prognostics methodologies for drilling operations are critical to detect the progress of tool wear during the cutting operations, so that worn tools can be replaced in time, ultimately achieve an effective and economical manufacturing.

Many models have been developed and different approaches to tool wear monitoring have been studied over the years. Neural networks were proposed for drill wear classification and measurement [2,3]; model based techniques were used to detect tool wear and breakage for machine tools [4]; dynamic force modeling method has been applied to build dynamic model of cutting forces in drilling [5] or to establish the thrust force and torque models for drilling [6] to detect tool wear; statistical approach was also used for tool wear analysis [7]; besides, from control system design standpoint, signals such as feed, speed and torque were investigated for drilling process control [8-10]. However, from the essence of prognostics, the remaining (effective) life for drilling remaining life estimation has not been discussed sufficiently [11-13]. Traditionally, the developed detection algorithms work well for a time-based tool life management. Recent manufacturing outsourcing business environment necessitates more focus on machine performance degradation to optimize the tool management for near-zero downtime and improved manufacturing performance. The unmet needs for drill monitoring are how to effectively predict its remaining life and manage the tool change to minimize downtime and costs.

In terms of signals, many sensing techniques of drilling process have been reported in the previously published literatures including touch sensors, power, acoustic emission, vibration, torque and/ or thrust force, and the vision systems [14]. Among these methods, vibration monitoring technique is a frequently used approach for drill failure detection. It has shown that vibrations are influenced by the torque and thrust force which are the major excitation sources in drilling [15].

This paper proposed a new approach to predict remaining life of cutting tool using logistic regression combined with ARMA model based on vibration signals collected during the drilling operation. LR analysis is used to evaluate health condition of a tool and ARMA model is used to predict future performance. The effectiveness of the method is further validated through an experiment using a twist drill on a steel work-piece.

II. TOOL WEAR IN DRILLING OPERATION

Tool wear in drilling is a progressive and comparatively slow phenomenon whereas tool failure and cutting edge breakage are usually catastrophic [16]. Even though a drill tool begins to wear as soon as it is placed into operation, the wear occurs at an accelerated rate once a drill becomes dull. As illustrated in Fig.1, drill wear states can be classified into different wear stages as a function of tool life [17]: 1. initial wear; 2. slight wear (regular stage of wear); 3. moderate wear (micro breakage stage of wear); 4. severe wear (fast wear stage); and 5 worn-out (or tool breakage).

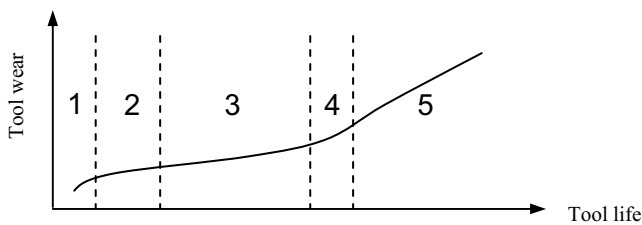


Fig.1 Tool wear evolution

It becomes possible that replace tool at proper time as long as the tool wear can be assessed and predicted proactively.

III. METHODOLOGY: A HYBRID PROGNOSTICS METHOD FOR CONDITION ASSESSMENT AND PREDICTION

In this paper, logistic regression is employed to perform drilling tool wear condition assessment based on the features extracted from vibration signals. Tool life prediction is performed using ARMA model. Wavelet Packet Decomposition is used to extract useful features from vibration signals.

A. Feature extraction using wavelet packet decomposition

Wavelet transform is one of the useful methods for analyzing the non-stationary signals [18]. Wavelet packets consist of a set of linearly combined usual wavelet functions. A wavelet packet function has three indices, $\Psi_{j,k}^n(t)$, where integers n, j and k are the modulation, the scale and the translation parameter, respectively,

$$\Psi_{j,k}^n(t) = 2^{j/2} \Psi^n(2^j t - k) \quad n=1,2,\dots \quad (1)$$

The wavelets Ψ^n are obtained from the following recursive relationships,

$$\Psi^{2n}(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} h(k) \Psi^n(2t - k) \quad (2)$$

$$\Psi^{2n+1}(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} g(k) \Psi^n(2t - k) \quad (3)$$

Note that the first wavelet is the so-called mother wavelet function,

$$\Psi^0(t) = \varphi(t), \quad \Psi^1(t) = \Psi(t) \quad (4)$$

The discrete filters $h(k)$ and $g(k)$ are quadrature mirror filter (QMF) associated with the scaling function $\varphi(t)$ and the mother wavelet function $\Psi(t)$. The decomposed wavelet packet component signal $f_j^n(t)$ can be expressed by a linear combination of wavelet packet functions $\Psi_{j,k}^n(t)$ as follows,

$$f_j^n(t) = \sum_{k=-\infty}^{\infty} c_{j,k}^n \Psi_{j,k}^n(t) \quad (5)$$

The wavelet packet coefficients $c_{j,k}^n$ can be obtained from,

$$c_{j,k}^n = \int_{-\infty}^{\infty} f(t) \Psi_{j,k}^n(t) dt \quad (6)$$

Usually, direct assessment from all wavelet packet coefficients often results in tedious or leads to inaccurate decisions. In our study, we define the wavelet packet node energy as

$$e_{i,j} = \sum_k (c_{j,k}^n)^2 \quad (7)$$

which measures the signal energy contained in some specific frequency band indexed by parameters i and j . In the following, each (i, j) is called as a wavelet packet node. For time-shifted signals, node energy representation provides a more robust signal feature than using coefficients directly. Therefore, each wavelet packet node energy value was defined as an individual feature component and was used as a robust exploration of the specific signal features that provide useful information for evaluation purposes.

B. Feature Selection using Fisher's criterion

Direct manipulation of a whole set of node energies is ineffective since the space normally has very high dimensionality, and the existence of undesired components makes the classification unnecessarily difficult. Feature selection in feature measurement space is to select the feature components that contain discriminant information and discard those feature components that provide little information. The feature components $\{f_k \mid k = 1, 2, \dots, n\}$ can be ranked as:

$$J(f_1) \geq J(f_2) \geq \dots \geq J(f_{n-1}) \geq J(f_n) \quad (8)$$

where $J(\bullet)$ is a criterion function for measuring the discriminant power of a specific feature component. The feature subset can be selected from the available features that have larger criterion function values. Fisher's criterion [19] was used as criterion function and is defined as:

$$J_{f_k}(i, j) = \frac{|\mu_{i,f_k} - \mu_{j,f_k}|^2}{\sigma_{i,f_k}^2 + \sigma_{j,f_k}^2} \quad (9)$$

where μ_{i,f_k}, μ_{j,f_k} are the mean values of the k -th feature, f_k , for class i and j , and $\sigma_{i,f_k}^2, \sigma_{j,f_k}^2$ are the variance of the k -th feature, f_k , for class i and j correspondingly. This will significantly simplify the design of the logistic regression model and enhance the generalization capability of performance assessment process.

C. Condition assessment using logistic regression

Usually, machine condition description from daily maintenance records/logs is a dichotomous problem (either normal or failure) which can be represented using a logistic regression function [20-22]. The goal of logistic regression is to find the best fitting model to describe the relationship between the categorical characteristic of dependent variable (the probability of an event occur, constrained between 0 and 1) and a set of independent variables, as indicated in Eqn. 10. The logistic function is:

$$\text{Prob(event)} = P(x) = \frac{1}{1 + e^{-g(x)}} = \frac{e^{g(x)}}{1 + e^{g(x)}} \quad (10)$$

Since $P(x)$ ranges between 0 (normal) and 1 (failure), the logistic function can be treated as a probability distribution function [21]. The S-shape of the function in Fig.2 indicates a relative low probability of occurrence until some threshold is reached, at which time the probability of occurrence increases rapidly [23].

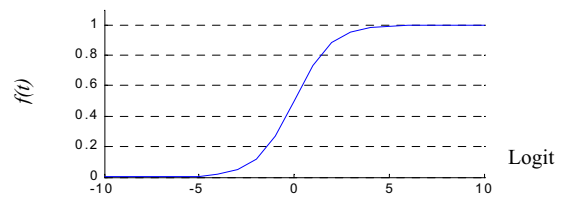


Fig.2 Logistic function

The logistic or logit model is:

$$g(\mathbf{x}) = \log \frac{P(\mathbf{x})}{1 - P(\mathbf{x})} \quad (11)$$

where $g(\mathbf{x})$ is a linear combination of the independent variables $x_1, x_2, \dots, x_k$, or

$$\begin{aligned} & \log [\text{Prob}(\text{event})/\text{Prob}(\text{no event})] \\ &= g(\mathbf{x}) = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \end{aligned} \quad (12)$$

The pre-condition for calculating $P(\mathbf{x})$ is determining parameters α and $\beta_1, \dots, \beta_k$ in advance. Due to the fact that the dichotomous dependent variable makes estimation using ordinary least squares inappropriate, rather than choosing parameters that minimize the sum of squared errors, estimation in logistic regression chooses parameters of α and $\beta_1, \dots, \beta_k$ using maximum likelihood method [23]. Then, the probability of failure for each input vector $\mathbf{x}$ can be calculated according to Eqn. 10.

D. Remaining life prediction using ARMA model

The essence of prognostics is the estimation of remaining life in meaningful terms that have consequence in the maintenance decision process. Fig. 3 indicates the schematic of predicted time to failure (TTF). At the moment of point 1, the corresponding prediction trajectory is shown as dash curve prediction 1, and TTF1 is time to failure of the moment. The prediction tendency and time to failure of point 2 are also shown in Fig.3. At each moment, predicted tendency and time to failure should be calculated dynamically.

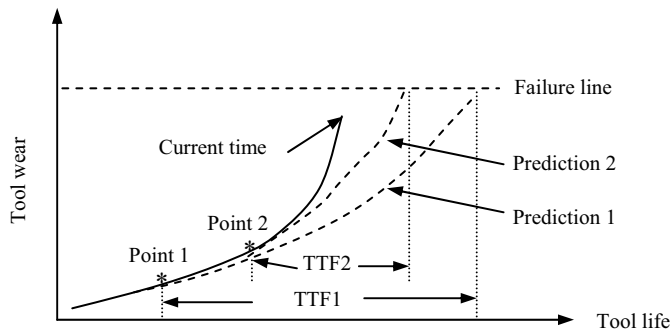


Fig.3 Predicted time to failure at each moment

In this paper, prediction of degraded performance is accomplished using ARMA (p,q) model by the degradation trend from the logistic regression module. The ARMA (p,q) model is one of the most traditional techniques in statistical time series analysis. The assumed model can be presented as:

$$y_t = \alpha_1 y_{t-1} + \dots + \alpha_p y_{t-p} + \varepsilon_t - \beta_1 \varepsilon_{t-1} - \dots - \beta_q \varepsilon_{t-q} \quad (13)$$

Where P is the order of the autoregressive part, q is the order of the moving – average part, $\alpha_1, \dots, \alpha_p$ are the autoregressive parameters and $\beta_1, \dots, \beta_q$ are the moving-average parameters. ε_t denotes the series of errors. The way of approaching the modeling of such an ARMA process is to first determine the model orders P and q . This part is done by off-line. The specific procedure to choose suitable p,q :

Do{

ARMA Model order (p,q) selection

ARMA Model validation

} while ARMA model is rejected;

After P and q are determined [24], on-line ARMA model parameters identification is implemented using real-life performance index, then prediction can be done based on the dynamic ARMA model.

The extrapolation of trends based on recent observations is a common method for calculating remaining life. This calculation alone, however, does not provide sufficient information to form a decision or corrective actions. Therefore, the failure risk range (distribution) is discussed in this study. Figure 4 illustrates a hypothetical remaining life probability density function (PDF) model.

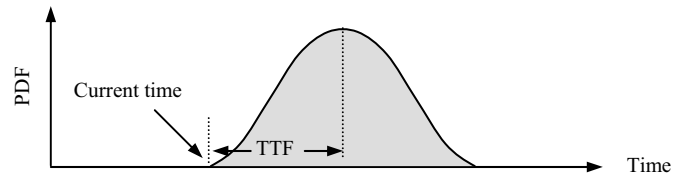


Fig. 4 Remaining life PDF model

Regardless of which PDF is used, the probability of failure is numerically equal to the area under the PDF function between any two points in time. As long as it holds true that the component will eventually fail and has not failed yet, the total area under each of these curves from the present to eternity must equal to 1.0. The expected point of failure is shown as the middle of the distribution, but the decision point is actually earlier and is a function of the shape of the PDF.

IV. EXPERIMENT AND ANALYSIS

The experimental setup and the corresponding machining information are shown in Fig. 5.

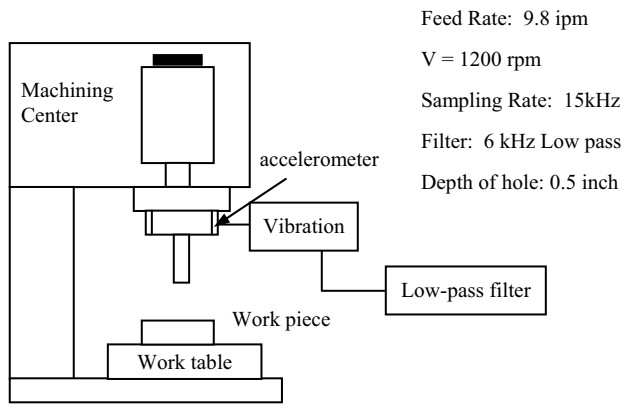


Fig.5 Schematic diagram of the experimental set-up

150 holes were drilled continuously on a work piece. After the drilling operations, tool wear appeared obviously. Here, total 20 (very beginning 10 and the last 10) holes' data was used for logistic regression model training. The specific procedures of the method are summarized as follows: 1. Seven-level WPD is generated for each vibration signal, in this case, Daubechies wavelet (DB4) was used for decomposition; 2. Energy based feature measures, as discussed in Section 3.1, are then calculated for each WPD of signal from step 1. This results in a 128-dimensional feature vector for each vibration signal; 3. Identify a subset of feature components using Fisher's criterion, as discussed in Section 3.2, to form an input vector for the logistic regression model.

Eventually, for each drilling operation, a 10 dimensions feature vector (packet energies from vibration) is input into LR model for training performance assessment LR model.

Based the training data of the 20 holes, 10 holes of sharp (good) condition ('0'-normal in Eqn.(10)) versus 10 holes of dull (fault) condition ('1'-fault, Eqn.(10)) were used to estimate the parameters α , β_1 to β_{10} using maximum likelihood method.

Then the trained LR model was used to assess the overall 150 holes condition. The assessment result is shown in Fig.6. The tool wear process from sharp to dull can be observed clearly. The whole process was also separated into 5 phases to match the tool wear evolution process shown demonstrated in Fig.1.

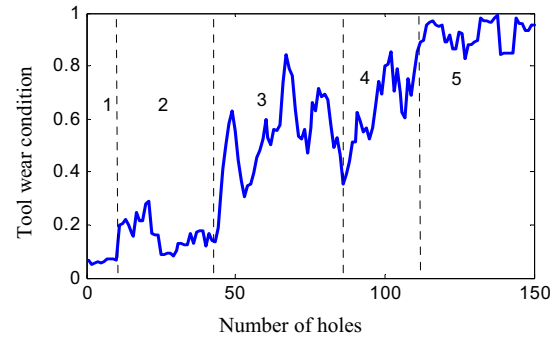


Fig. 6 Assessment results of all 150 holes

The index 'time to failure' (TTF) can be represented as holes to failure. The trajectories of two TTF predictions are shown in red line in Figure 7, where predictions were conducted at the 105th hole and the 122nd hole. The number of holes to failure at the 105th hole is 38 and the number of holes to failure at the 122nd hole is 14.

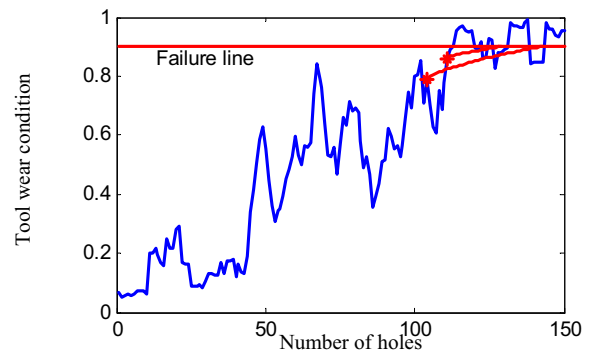


Fig.7 TTF prediction

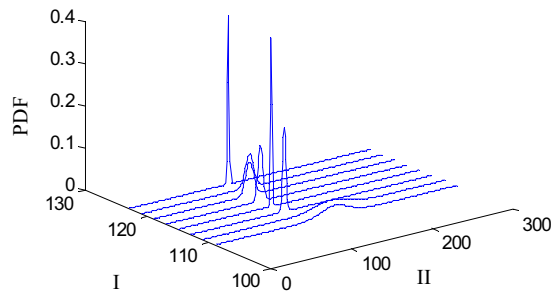


Fig.8 PDF of predicted TTF

I: Number of holes; II: Predicted number of holes

PDFs are also used for failure prediction as illustrated in Figure 8. If the hole number is more than 115, the corresponding PDF shapes getting narrower. It shows a wear condition progresses toward dull and the probability of worn-out is very high. The tool was almost worn-out after 130 operations.

V. CONCLUSIONS

This paper presented a practical and reliable hybrid method for detecting tool wear and predicting tool remaining life, which will assist operators in just-in-time tool replacement and machine maintenance for near-zero downtime machining performance. The effectiveness of the hybrid prognostic method has been validated in a drilling operation. The results of tool performance assessment and TTF prediction are promising. Note that the methodology is generic which could be applied to diverse industrial process to prognose machinery performance.

ACKNOWLEDGMENT

The data used in this paper was acquired from S.M.Wu Laboratory at the University of Michigan- Ann Arbor.

This research work is also funded by HLJNSF (E200634) and 9140A17030606HT0171.

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